

DIYA NILESH DESAI

+91 9737638128 | desaidiya041@gmail.com | linkedin.com/in/diya-desai-080339311 | github.com/diyadesai-cell

PROFESSIONAL SUMMARY

B.Sc. Data Science graduate and AI/ML Intern building production machine learning systems: a recommendation microservice (Item2Vec, vector search, re-ranking) for a live food-delivery app, plus RAG, LLM and computer vision projects. Also experienced in data analysis and KPI dashboards. Seeking an entry-level AI/ML or Data Science role.

TECHNICAL SKILLS

Programming: Python, SQL, R, Git/GitHub

Machine Learning: Item2Vec, Recommender Systems, Market-Basket Analysis, Vector Similarity Search, Re-ranking, Scikit-learn, Regression/Classification, Time-Series Forecasting (ARIMA, Holt-Winters)

GenAI & LLMs: RAG, LangChain, HuggingFace, ChromaDB, Groq/Llama 3.3, Quantized GGUF Inference (llama-cpp-python), Qwen2.5, Diffusion/FLUX Models

Deep Learning & CV: YOLOv8, OpenCV, CNN, Image Annotation

MLOps & Backend: FastAPI Microservices, Offline Batch Pipelines, ETL, Versioned Model Publishing, Load Testing, GPU Memory Management

Data & Analytics: PostgreSQL (pgvector), MySQL, Power BI, Grafana, Excel, Plotly, EDA, Data Validation, RapidFuzz

EXPERIENCE

AI/ML Intern | Easy Cater — July 2026 – Present

- Designed and built **ec-recommendation**, a production recommendation microservice (FastAPI, PostgreSQL/pgvector) serving personalized “you might also like” suggestions in the Easy Cater mobile app.
- Trained an Item2Vec model on order-basket data as an offline nightly job, published via atomic-swap versioned vector indexes; built a tiered fallback chain (Item2Vec → order history → popularity) for cold-start users under concurrent load.
- Built a Cart Recommendation Engine for in-cart cross-selling using cart-item embedding centroids, with category-aware re-ranking across ~178K menu items and ~4,750 categories.
- Automated menu-item categorization with a 3-tier pipeline (exact match → RapidFuzz → quantized Qwen2.5-0.5B via llama-cpp-python) integrated into the ETL job, removing manual labeling.
- Added day-part (meal-slot) awareness via a trained classifier enforced as a SQL predicate across all recommendation tiers; validated on an isolated test database before production.
- Benchmarked GPU memory and capacity (Qwen-Image-Edit vs FLUX.1[dev]) to inform model selection for ec-imagegen, an AI photo enhancement pipeline for merchant food photos.
- Data analysis:** built an 8-module Grafana dashboard (50+ KPIs) on MySQL; delivered click, peak-hour, search-demand and delivery SLA analyses for leadership; ran a 3-tier data validation process before production.

AI/ML PROJECTS

Petal AI — Document Q&A Study Assistant | Python, FastAPI, LangChain, HuggingFace, ChromaDB, Groq LLM

- Full-stack RAG application for conversational Q&A over any uploaded PDF, with LangChain memory and Groq Llama 3.3 70B for context-grounded answers. (github.com/diyadesai-cell/petal-ai)

Smart Parking System — Computer Vision | Python, YOLOv8, OpenCV | Hackathon 2nd Runner-Up

- Trained a YOLOv8 model to classify parking occupancy; built the full pipeline from annotation to evaluation. 2nd Runner-Up, Indigenous Hackathon, March 2025. (github.com/diyadesai-cell/Smart_parking_system)

Driver Drowsiness Detection | Python, CNN, Deep Learning

- CNN with Eye Aspect Ratio and facial landmarks to classify driver alertness on live video.

Renewable Energy Forecasting | ARIMA, Holt-Winters, Streamlit

- State-wise solar and wind capacity forecasts over a 2–3 year horizon with an interactive Streamlit dashboard.

Flight Delay Analysis & Prediction | Python, Scikit-learn, Plotly

- Analyzed 539,383 flights (44.5% delayed); built a Logistic Regression model to predict delays from airline, day and flight length.

DATA ANALYSIS PROJECTS

Nutritional Health Dashboard | Power BI, Excel, DAX

- Cleaned multi-source retail and public nutrition datasets and built Power BI dashboards with custom DAX measures for nutrient KPIs and brand/category insights.

Easy Cater Operations Dashboard | Grafana, MySQL

- 8-module dashboard with 50+ KPIs covering customers, restaurants, riders, orders, revenue, menu, retention and live alerts.

EDUCATION

B.Sc. Data Science | Navrachana University, Vadodara | CGPA: 8.16 / 10 | May 2026

CERTIFICATIONS

AWS Academy – Cloud Foundations, Data Engineering, Machine Learning Foundations | AWS – ML for Natural Language Processing | Tata – Data Visualization | Deloitte Australia – Data Analytics Job Simulation | IBM – Getting Started with Data